# Assignment 02

## Which Aviation Accidents Turn Fatal?

Teaching Week 08 | Individual work. Submit in R or Python.

**Purpose**

Assignment 1 left you with one documented row per Canadian air occurrence. This assignment adds the aircraft involved and asks a predictive question of that table, namely which accidents turn fatal. About one accident in ten is fatal, so a model that never predicts a fatality already looks accurate. You will compare a baseline, linear discriminant analysis, $k$-nearest neighbours and two support vector machines on the same frozen folds, check the estimate against time order and against the luck of tuning, and test whether the best model is genuinely better than its closest rival. The marks fall on the protocol and the argument, not on the highest score.

## Learning Outcomes

After completing this assignment, you should be able to

- extend an analytical table with a second published table while keeping one row per occurrence;
- define a rare binary target, choose predictors known before the outcome, and explain why a column that describes the outcome leaks;
- tune $k$NN and a support vector machine inside cross-validation on folds frozen once and shared by every model;
- judge a classifier of a rare class by balanced accuracy, its confusion matrix, and the precision and recall of that class;
- check an estimate against time order and against the luck of a search, and test a gap with the corrected resampled $t$-test; and
- publish a reproducible analysis that another analyst can rerun from the submitted materials.

## The Dataset

Everyone extends the record used in Assignment 1. The occurrence table already holds the target, and the aircraft table supplies most of the predictors.

**The Aircraft Table of the Air Occurrence Record**

**Publisher.** Transportation Safety Board of Canada, from its Aviation Safety Information System
**Official record.** `https://www.tsb.gc.ca/eng/stats/aviation/data-5.html`
**Licence and format.** Open Government Licence – Canada; one CSV table, described by the same data dictionary as the occurrence table
**Observational unit.** One aircraft in one occurrence. The table joins the occurrence table on `OccID`, and an occurrence with several aircraft has several rows.
**Variables to locate in the dictionary.** maximum take-off weight; year of manufacture; number of engines; aircraft type; operator type; damage level; and, in the occurrence table, the date, region, latitude, longitude, occurrence type, country and total fatal count
**Use in this assignment.** Add aircraft predictors to your Assignment 1 table and predict whether a Canadian accident was fatal. The record carries no flight hours, so a predicted probability is not a risk per flight.

## Retrieving the Data

Download the aircraft table, a fresh copy of the occurrence table and the dictionary on the same day, so that the keys of both tables describe one snapshot, and run your Assignment 1 import on the fresh occurrence table. The site refuses a request that does not look like a browser, and a plain scripted download returns HTTP 403. Use a browser download, or set a browser-like user agent in your command and record it.

- **Dataset page.** `https://www.tsb.gc.ca/eng/stats/aviation/data-5.html`
- **Aircraft table.** `ASISdb_MDOTW_VW_AIRCRAFT_PUBLIC.csv`
- **Occurrence table.** `ASISdb_MDOTW_VW_OCCURRENCE_PUBLIC.csv`
- **Data dictionary.** `ASISdb-dd.csv`

Record the access date, byte size and SHA-256 checksum of every file, as in Assignment 1, and keep the original files privately. If a resource is unavailable, retain the error and access time and contact the instructor for a documented course copy.

## Data Contract

Your modelling table must satisfy every rule below before you fit anything.

- Keep Canadian accidents from 1 January 2000 onward, using the dictionary's country and occurrence type codes. The occurrence table can repeat an `OccID` on several rows, so reduce it to one row per occurrence first, and report the row count after each step.
- Keep the accidents that involve exactly one aircraft, meaning one distinct `AcID`. The aircraft table can hold several rows for that one aircraft, so collapse them to one row per aircraft before joining, and check that the predictors you take from it agree across those rows.
- Hold one row per occurrence, keyed by `OccID`, with no duplicate key.

- Define the target as fatal when `TotalFatalCount` is above zero, and report its class counts.
- Choose at least six predictors, and only these, among the aircraft's take-off weight on a log scale, its age at the accident, its number of engines, the latitude, the longitude, the month, the operator type, the aircraft type and the region. The latitude and the longitude count as two. The dictionary stores the longitude as a positive number with its hemisphere in a separate field, so check its sign before using it. Merge rare categories with a rule you state, such as fewer than 100 accidents, applied the same way to every fold.
- Use no predictor that describes what happened. Damage level is forbidden, and so are every injury count, the injury feature you joined in Assignment 1, the occurrence classification, the ICAO occurrence category, which names the kind of event, and the summary text.
- Do not use weather, light or temperature. Each is recorded for only about one accident in nine.
- Keep at least 5 000 complete rows, and state the rule for every row you drop and every gap you fill.
- Keep the occurrence date of every row for the time-ordered check, even if only the month is a predictor.

**What the Contract Does Not Settle**

Meeting the contract shows that your table is faithful to the published record and free of the obvious leaks. It does not show that your folds are honest, that your comparison is fair, or that a model fitted on earlier years would still work on the accidents of next year.

## Modelling Protocol

### One Set of Folds for Every Model

Cut the rows into five stratified outer folds, and inside each outer training set cut five stratified inner folds. Freeze the folds once, in a file or an object your code reads, and use them for every model, so every comparison is paired. Choose every setting by its mean balanced accuracy on the inner folds, never by plain accuracy, and score each model once on the outer fold it never saw. Put scaling, indicator columns and any imputation inside the pipeline, so they are refitted on every training set.

### The Models

Fit a baseline that always predicts the most common class; linear discriminant analysis twice, once with the class shares of the training rows as its priors and once with equal priors, both fixed before you see any score (the `prior` argument of `MASS::lda` in R, `priors` in scikit-learn); $k$NN with $K$ chosen inside; a radial support vector machine with $C$ and $\gamma$ chosen inside on a grid of powers of ten; and the same support vector machine with balanced class weights, tuned the same way. Report the grid you searched for each model. If a chosen setting sits on the edge of its grid, widen the grid where you can, and say so where you cannot, as for $K = 1$.

### A Metric for a Rare Class

Report balanced accuracy, the mean of the recall of the fatal and the non-fatal class, as the headline metric for every model. For the model you recommend, add its confusion matrix and the precision and recall of the fatal class. Plain accuracy may be reported beside balanced accuracy, never instead of it.

**Modest Gains Are the Honest Result**

The aircraft, the place and the season say little about whether a crash will kill. On a recent snapshot the best of these models reached a balanced accuracy of roughly 0.56 to 0.59, against 0.50 for the baseline. A score in that range, honestly estimated and honestly compared, can earn full marks, because the marks fall on the protocol and the argument rather than on the score. A much higher score is more likely a leak than a discovery, so check it before reporting it.

## Reproducible Workflow

You may complete and submit the assignment in either R or Python. Both paths are marked against the same evidence and the same criteria, and both must run end to end from one documented entry point in your portfolio repository.

Use the methods taught in the course up to and including support vector machines; decision trees and later methods are not part of this assignment. Write your own preparation, fitting, evaluation and reporting source, and generate every reported number, table and figure from it. Record the software, package versions, random seeds and the exact command that reproduces the analysis.

## Work Programme

### 1. Register the Source and Ask the Question

Record the provenance the contract requires. State the target, the unit of prediction, and a decision a working model could inform, in one sentence each.

### 2. Join the Aircraft Table

Use the dictionary to establish the key and the cardinality between the two tables before joining. Report how many occurrences have no aircraft row, one, or several, the row counts before and after the join, and unmatched keys in both directions. The dictionary warns that take-off weight may be in another unit before 2016, so look at its distribution by year before trusting the log scale, and treat an impossible aircraft age as missing with a stated rule.

### 3. Build the Modelling Table and Rule Out Leakage

Publish a variable table giving each kept column its type, its unit, its role and its share of missing values, and report the class counts of the target. In one paragraph, explain why damage level would make the models look better than they could ever be in use, and name the moment at which each predictor you kept is known.

### 4. Fix the Folds

Create the outer and inner folds, report the number of fatal accidents in each outer fold, and show that the same folds reach every model.

### 5. Baseline, LDA and $k$NN

Explain why the baseline scores 0.5 in balanced accuracy whatever its plain accuracy. Compare the two LDA fits and explain what the priors do to a class that makes up a tenth of the rows. Report the

inner scores of $k$NN across $K$ and the outer score of the chosen $K$, and say what a majority vote among neighbours does when most neighbours are non-fatal.

### 6. The Support Vector Machine

Tune $C$ and $\gamma$ inside, with the predictors scaled inside the pipeline. Show the inner grid for one outer fold and explain what its rows, columns and diagonals mean. Then tune the version with balanced class weights and compare the two on the recall and the precision of the fatal class as well as on balanced accuracy.

### 7. Is the Estimate Honest?

For every tuned model, compare the best inner score its search reported with its outer score, and explain any gap as the luck of the search. A gap close to zero, or one whose sign changes from fold to fold, is a valid finding, and it says the search had too little to choose between. Then train on the accidents up to the end of 2018 and test on every accident from 2019 onward, tuning inside the earlier years with stratified inner folds, for the baseline, the better LDA and the weighted support vector machine. State the rows and fatal accidents on each side. A test set with about 150 fatal accidents moves balanced accuracy by several hundredths with the luck of which accidents fall in it, so say whether the change is larger than that before you explain it. Say which estimate answers the question a user of your model would ask.

### 8. Compare and Conclude

Tabulate the outer score of every model on every fold, with the mean. Test the gap between your best model and its closest rival with the corrected resampled $t$-test, stating the overlap term $n_{test}/n_{train}$ your folds imply. Recommend one model, show its confusion matrix and the precision and recall of the fatal class, describe the accidents it misses, and state what the evidence does not show.

## What to Submit and Where

Keep the work under `assignments/assignment-02/` in a private work repository with instructor access until the deadline has passed and the instructor opens sharing. Commit the finished report, source, and evidence first. In a second commit, record that completed-work commit's full SHA and immutable commit URL in the directory's `README.md`. Push both commits to the default branch by the deadline. A commit cannot contain its own SHA; the marker assesses the snapshot named in the pushed README. The assignment is released on October 27 and is due at 23:59 on November 17, in the course time zone.

Include

- a concise PDF report carrying the question, the join and its counts, the variable table, the leakage argument, the protocol, the results for every model, the honesty checks, the comparison and test, and the limitations;
- the complete R or Python source, plus every table, metric and figure the report cites, written by that source rather than edited by hand;
- the frozen fold file, or the code that creates it together with its seed;
- a README carrying the submitted commit SHA and its commit URL, the official data URLs, the retrieval steps, the access date, byte size and SHA-256 checksum of each file, the exact execution command, and the runtime environment; and
- no redistributed copy of the source data, which is linked rather than committed.

A submission is complete only when the required files open and the analysis can be reproduced from the submitted materials. Late work loses 5 percentage points of the available assignment mark for each late calendar day, for up to seven calendar days, subject to the syllabus's accommodation provisions. Cite the dataset and every non-trivial external source, and submit only your own work.

When sharing opens, publish the submitted commit with its existing history for discussion at the next class meeting, if one follows, and in your portfolio. Make corrections in new commits. Do not amend, rebase, squash, force-push, delete, or backdate submitted history. The pushed README record establishes what was submitted and when; Git timestamps alone do not prove timely completion.

Generative AI output is not authorized for this assignment; the syllabus's academic integrity and private-repository exception provisions apply.

## Grading

This assignment is worth 7.5 percent of the course grade. The weights below are percentage points of that course grade and sum to the assignment's weight.

**Table 1.** Assignment 02 marking criteria.

| Criterion | Course weight (%) |
|---|---|
| Provenance, join and modelling table. Aircraft join checked with counts, filters and gaps stated, predictors justified, leakage of damage level explained. | 1.0 |
| Fold design and leakage control. Frozen stratified folds shared by every model, every learned step fitted inside the training rows. | 1.5 |
| The models. Baseline, both LDA fits, kNN and both SVMs tuned inside by balanced accuracy with grids reported, SVM grid read, priors and class weights explained. | 1.5 |
| The honesty checks. Lucky-search gap reported for every tuned model, time-ordered check with a stated cut and its change explained. | 1.5 |
| Comparison and conclusion. Paired fold table, corrected test with its overlap term, confusion matrix and fatal-class precision and recall read, limits stated. | 1.5 |
| Reproducibility and communication. Runs end to end from one entry point, artefacts generated, report clear and cited. | 0.5 |
| Total | 7.5 |

## How This Relates to the Weekly Labs

This assignment assesses predictive modelling on one record, carried from the first prediction models to the support vector machine. Lab 08 practises the support vector machine workflow on a different, multiclass ecological record, so the two do not ask for the same study, and work from Lab 08 does not satisfy this assignment. The best nine of the ten weekly labs count toward the lab component.